

ELA-1 PROJECT REPORT

Track 1: Phishing & Malicious Email Classifier

Deep Learning Applications — Department of CSE-CyS, DS and AI&DS;

Abstract

This project develops a deep-learning pipeline for classifying raw textual email payloads into Legitimate and Phishing/Malicious categories. A synthetic corpus of 2,000 labeled emails was created, exceeding the assignment requirement of 1,500 samples. The pipeline performs text normalization, tokenisation, vocabulary construction, integer encoding and fixed-length padding/truncation. A bidirectional Long Short-Term Memory (BiLSTM) network performs sequence classification, followed by an inference function that simulates Gmail-style email classification.

Dataset & Data Preprocessing

The corpus contains 1,000 base legitimate and 1,000 base phishing/malicious examples generated from multiple templates. Legitimate messages include project updates, meetings, invoices and reports. Malicious messages include credential requests, account verification, payment failures, rewards and suspension warnings. A 5% label-noise injection makes the synthetic benchmark less trivial. Text is lowercased, non-alphanumeric characters are normalized to spaces, tokenised on whitespace, mapped to a training-only vocabulary capped at 5,000 tokens, and padded/truncated to 60 tokens.

Model Pipeline

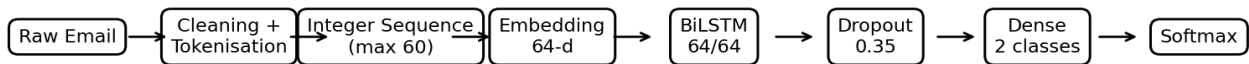


Figure 1. End-to-end architecture.

MODEL ARCHITECTURE & TRAINING

Architecture

The model uses a 64-dimensional embedding layer followed by a bidirectional LSTM with 64 hidden units per direction. The final forward and backward hidden states are concatenated, passed through dropout (0.35), and mapped to two output classes. Cross-entropy loss and Adam optimization are used.

Hyperparameters

Parameter	Choice
Dataset	2,000 synthetic emails
Split	70% train / 10% validation / 20% test
Vocabulary	5,000 most frequent training tokens
Sequence length	60 tokens
Embedding	64 dimensions
BiLSTM	64 units per direction
Dropout	0.35
Optimizer	Adam
Learning rate	0.001
Batch size	64 train / 128 eval
Epochs	12

Training Curves

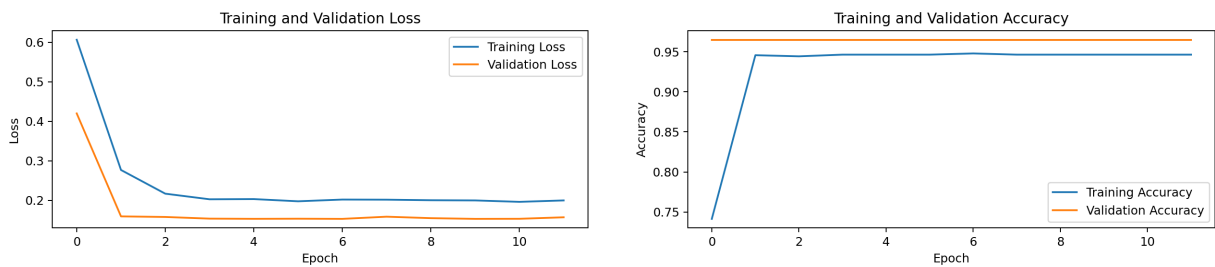


Figure 2. Training/validation loss and accuracy curves.

EVALUATION & CONCLUSION

Test-set Results

Metric	Score
Accuracy	95.50%
Precision	94.71%
Recall	96.57%
F1-score	95.63%

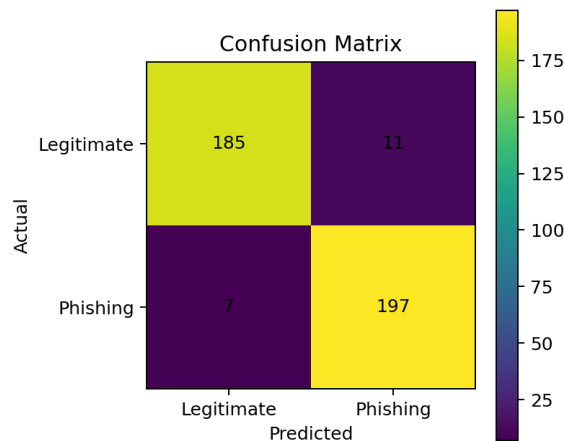


Figure 3. Confusion matrix: TN=185, FP=11, FN=7, TP=197.

Discussion

The BiLSTM achieved 95.50% accuracy and an F1-score of 95.63% on the held-out synthetic test set. Recall of 96.57% is useful for the security track because it reflects the proportion of phishing examples detected. However, the result must not be interpreted as real-world phishing-detection performance because the corpus is synthetic and template-driven. It does not reproduce the diversity of real mail, HTML, attachments, multilingual content, URL obfuscation or adversarial wording.

Conclusion & Future Work

The implementation satisfies the ELA-1 requirements with a reproducible sequential LSTM model, preprocessing workflow, architecture, loss/accuracy curves, confusion matrix and F1-score. Future work should augment the synthetic corpus with real public email/phishing datasets, test on an unseen external dataset, address class imbalance, and expose the trained classifier through a Flask/FastAPI endpoint.

Reproducibility: The supplied notebook uses seed 42 and regenerates the dataset, preprocessing, model and evaluation.